

The Utility Problem

A transformer load refers to the amount of power drawn by the electrical load connected to a transformer. For grid operators, it is important to ensure that a transformer is not overloaded, as this can damage the transformer and potentially lead to power outages or other issues. Overheating and, consequently, reduced transformer lifespan are some of the consequences of an overloaded transformer.

With an increasing number of new connection requests in the distribution grid, such as businesses, households, EV chargers, heat pumps, etc., it is important to have a capacity management plan in place to ensure that the distribution grid and the transformers within it can accommodate the new connections without compromising the reliability and quality of service for existing customers. It is, therefore, necessary to assess the load changes for these new connection requests.

Grid operators traditionally use approximate load or load calculated from the reserved capacity (Master Circuit Breaker value) for estimating the transformer load of connected customers. With both approaches, this calculation will not take into account reality. Still, it will only be an estimate, which is also likely very inaccurate because the calculation will not include real measured data. To make matters worse, in the majority of the cases, there is no measurement at the transformer; only individual meters "behind" the transformer are measured.

For these reasons, a way of aggregating the transformer downstream load is necessary to model the transformer loading and consequently increase the transformer's expected loading for the new connection request load. This process will ensure that grid operators can make educated decisions about allowing new connections into the grid.

The Data Solutions and Methods

This use case will significantly expedite the work of a grid operator analyst who would otherwise need to search for data from various sources, connect them, and make corrections. However, with the Awesense Platform, the analyst has access to all the cleaned and synchronized data in one place, allowing immediate workability.

For conducting such an analysis, it is necessary to have several datasets ready. These datasets are as follows:

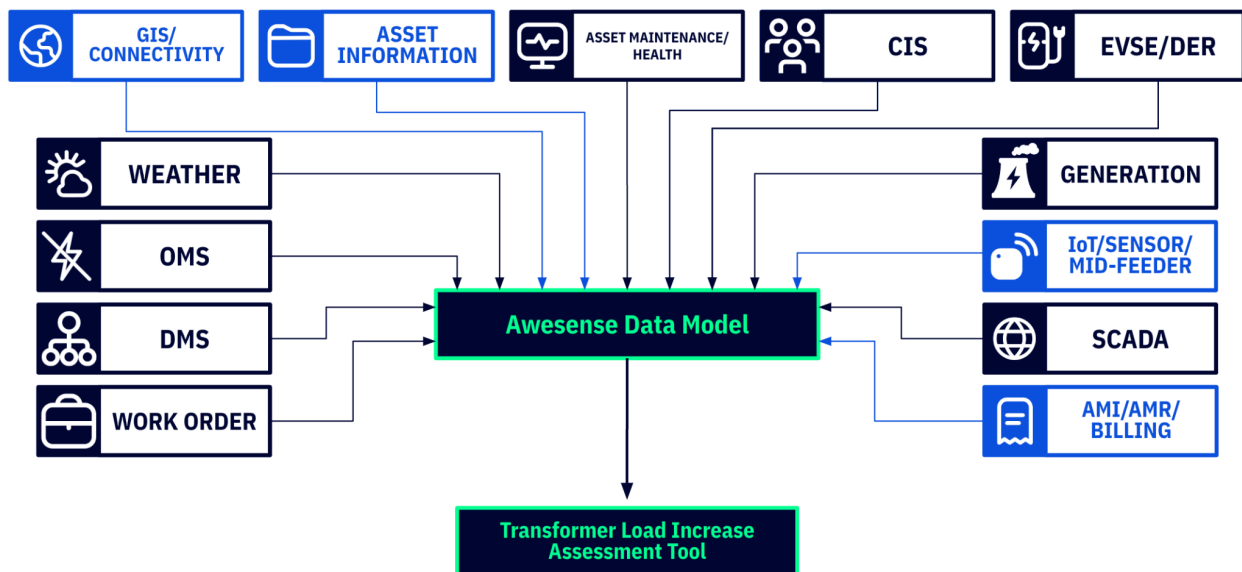
1. Transformer asset information (properties of the transformer), specifically the transformer capacity - kVA rating.
2. Knowledge of the connected consumption points (loads) to the transformer - this information can be obtained very easily using the trace function - `grid_get_downstream()`. *This function uses real grid connectivity data.*
3. Knowledge about the current load:

- a. at the transformer (load time-series data) - if the measurement is at the transformer
- b. or of the connected consumption points (load time-series data) - for each consumption point according to point 2, consumption data can be obtained using the function `ts_data_source_select()`.
4. Knowledge about the future load - typical load profiles of consumption points (time-series data) - this information can be obtained from the use case “Standard Consumption Patterns Creation”.

The calculation compares the total consumption behind the transformer with the future added consumption. These values are compared to the maximum capacity of the transformer for each time period. If the capacity is exceeded, a new load will cause the transformer to overload.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. GitHub [link](#) to Jupyter notebook.